Moving Object Detection using Median Background Modeling and Morphological Dilation

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Abstract

The reliable detection of moving objects is an essential requirement for modern video surveillance applications. This report details the implementation of a custom object detection pipeline using background subtraction, manual thresholding, and morphological operations. By calculating a static background model from a median of randomly sampled frames, the system effectively isolates moving entities in a static environment.

1 Introduction

The technology of motion detection has become one of the most important research areas in computer vision[cite: 11]. It serves as the foundation for a multitude of applications in video surveillance, ranging from indoor and outdoor security monitoring to traffic control and spot behavior detection[cite: 12, 23, 24]. Reliable detection of moving objects is an important requirement for such applications[cite: 25].

Among various approaches, background subtraction is the commonly used class of technique to detect moving regions in an image sequence[cite: 27]. This method isolates objects by comparing the current frame to a reference frame, making it highly dependent on a good background model to reduce the influence of dynamic scenes[cite: 28]. This report evaluates a custom approach using median-based static background modeling coupled with morphological dilation to extract foreground objects.

2 Methodology

The proposed pipeline consists of background initialization, frame differencing, and morphological refinement.

2.1 Background Initialization

To create a robust static reference, the algorithm initializes a background model by analyzing the first 100 frames of the video sequence. Rather than using a single frame, the system randomly samples 50 frames from this initialization period. A median filter is applied across the temporal axis of these sampled frames to compute a static background image. This median approach effectively removes transient moving objects from the reference scene. The resulting background frame is then converted to grayscale and smoothed using a Gaussian Blur with a kernel size of 21×21 to mitigate sensor noise.

2.2 Frame Differencing and Thresholding

As new frames are read, they are resized to a standard dimension and subjected to the same grayscale and Gaussian blur preprocessing. The algorithm then computes the absolute difference between the incoming blurred frame and the median background model. To binarize the image and isolate the moving regions, a fixed threshold (value = 30) is applied to the difference image.

2.3 Morphological Refinement

Background subtraction often yields noisy or fragmented foreground masks. To address this, morphological operations are applied[cite: 460]. Specifically, a dilation operation is executed over 5 iterations. Dilation expands the bright areas (foreground pixels) of the image by convoluting the image with a kernel, replacing the output pixel with the maximal pixel value overlapped by the kernel[cite: 464, 466, 468]. This process successfully fills structural holes inside the moving objects and bridges gaps caused by imperfect thresholding.

2.4 Contour Detection

Finally, the algorithm applies a contour discovery process to the dilated binary mask. To filter out residual noise, only contours possessing a spatial area strictly greater than 500 pixels are accepted. Bounding contours are then drawn over the original frame to highlight the detected entities.

3 Conclusion

The median-based background subtraction method combined with absolute differencing and morphological dilation provides a straightforward yet computationally efficient technique for object detection. While highly effective in strictly static environments, this baseline method establishes the foundational concepts necessary for understanding more complex, dynamic background subtraction models.